BRENMARK TELECOM

BSS/OSS Modernization Program

Enterprise Architecture Definition Document

A TOGAF Architecture Development Method (ADM) Case Study

Preliminary Phase through Phase H — Full ADM Cycle

Prepared by: Enterprise Architecture Function

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Table of Contents

1. Executive Summary

Brenmark Telecom cannot sell a single product that combines mobile, broadband, and (soon) 5G network slicing, because its billing, CRM, and network order-management systems each keep their own incompatible view of "customer" and "product." Provisioning a new service takes 3–7 business days because staff manually coordinate across three disconnected systems, and the Board has mandated 5G network-slicing monetization within 18 months — a capability none of the current systems can represent, sell, bill, or activate in any form today.

The Enterprise Architecture function recommends a "strangler fig" modernization strategy: a TM Forum Open API–conformant gateway and a new Product Catalog and Order Orchestration layer are placed in front of the legacy billing, CRM, and network inventory systems. Legacy systems — most importantly the billing engine's real-time charging and financial ledger, which serves approximately 6 million subscribers and is regulator-audited — keep running unchanged; new products, including converged bundles and 5G slices, are built on the new layer from day one, and capability migrates off legacy systems incrementally, governed by named transition architectures, rather than through a single risky cutover. A full BSS/OSS rip-and-replace and a disconnected greenfield-only build for new products were both evaluated and rejected (ADR-001): the former is incompatible with the 18-month deadline and concentrates all migration risk into one event, and the latter would permanently strand the existing 6 million subscribers on the very estate the program exists to fix.

The program is estimated at $18.4M in capital investment over 18 months, plus roughly 15% ongoing operational overhead on affected systems for approximately 24 months while old and new systems run in parallel. Against this, the program projects approximately $27.4M in lower 3-year total cost of ownership versus doing nothing ($53.4M as-is versus $26.0M to-be), plus approximately $23.3M in new converged-bundle and 5G-slice revenue over three years — a net 3-year benefit of approximately $50.7M relative to as-is, with a payback period of approximately 22 months from program start. The largest single benefit driver is new revenue, not cost avoidance: this is fundamentally a growth investment.

The migration runs across four waves over 18 months (following a Wave 0 governance and vendor-selection foundation), sequenced by technical dependency and gap priority rather than convenience — legacy adapters and the API gateway are built first in Wave 1 because nearly every other solution building block depends on being able to reach billing and network inventory through a stable API contract. Converged bundles and the first 5G-slice product launch in Wave 3, directly fulfilling the board's mandate, with Wave 4 dedicated to hardening, RBAC uplift, and governance closeout. This document is the consolidated Architecture Definition Document for the program, compiled across every phase of the TOGAF Architecture Development Method — Preliminary Phase governance setup through Phase H change management — and is intended as a single, self-contained reference of the architecture, its rationale, and its governance.

2. Preliminary Phase — Governance & Principles

Before Phase A vision work began, Brenmark's architecture function stood up a governance model sized for a program that must hit a board-mandated deadline without destabilizing a regulator-audited, real-time charging system serving 6 million subscribers — a governance model that must have real teeth over the highest-risk decisions (data ownership, security) while staying out of the way of routine engineering.

2.1 Architecture Principles

Twelve architecture principles were ratified by the Architecture Review Board (ARB) and govern every subsequent decision in this program, each written in standard TOGAF Name / Statement / Rationale / Implications format. Each ADR and architecture contract in this document cites these principles by number.

P-01 — Customer, Product, and Order Are Enterprise Data, Not System Data

Statement: Customer, product catalog, and order data are enterprise assets owned by the business, not by any single application. No system may be the sole, ungoverned source of truth for an enterprise data domain.

Rationale: Brenmark's core problem — the inability to launch converged bundles — exists precisely because billing, CRM, and order management each grew their own private notion of "customer" and "product." Treating these domains as enterprise assets with a defined system of record (Phase C) is the precondition for any converged product.

Implications: Every new system must consume, not duplicate, the designated system of record for a domain via governed APIs. Legacy systems that currently own duplicate copies must migrate to a subscriber/consumer role over the transition period defined in Phase F. New local caches require an explicit reconciliation strategy (see ADR-004).

P-02 — Interoperability Follows TM Forum Open API Standards

Statement: All new and modernized BSS/OSS interfaces expose and consume TM Forum Open APIs (e.g., patterns aligned to TMF622 Product Ordering, TMF637 Product Inventory, TMF629 Customer Management) rather than proprietary or point-to-point integration contracts.

Rationale: Telecom-specific standards exist precisely to solve the integration fragmentation Brenmark suffers from, and they give Brenmark a market of interchangeable vendors and integrators instead of lock-in to one proprietary integration style.

Implications: Every new integration point goes through the API gateway layer (Phase D) rather than direct point-to-point calls. Vendor selection (Phase E) weights Open API conformance heavily. Existing point-to-point integrations must be inventoried and scheduled for retirement.

P-03 — Modernize Incrementally; Never Freeze Revenue-Critical Systems

Statement: The program will not require a "big bang" cutover of any revenue-critical system (billing, active provisioning). Modernization proceeds in incrementally shippable, independently valuable waves.

Rationale: Brenmark's billing system processes real-time charging for 6M subscribers; a failed big-bang cutover is a business continuity risk the board will not accept, and the 18-month 5G-slicing deadline does not allow time to recover from one.

Implications: This principle directly drives the strangler-fig pattern selection over rip-and-replace (see ADR-001). It requires the program to accept a period of dual-write/dual-read complexity (ADR-004) as the cost of avoiding cutover risk.

P-04 — Architecture Decisions Are Traceable and Reversible by Default

Statement: Every architecturally significant decision is recorded as an ADR with alternatives considered and consequences stated, and wherever technically feasible, the architecture favors reversible decisions over irreversible ones.

Rationale: An 18-month program under board scrutiny cannot afford decisions whose rationale is lost to attrition, nor commitments that can't be unwound if a vendor or approach underperforms.

Implications: Solution building blocks are evaluated partly on exit cost (Phase E). ARB will not approve a design that lacks a documented rollback or reversal path for its highest-risk elements.

P-05 — Buy Commodity Capability, Build Differentiated Capability

Statement: Where a TM Forum ODA-aligned commercial product exists that meets Brenmark's functional and non-functional requirements, Brenmark buys and configures it rather than building bespoke software. Brenmark builds only where a capability is a genuine competitive differentiator (e.g., converged bundle pricing logic, 5G-slice monetization rules).

Rationale: Engineering effort is Brenmark's scarcest resource against an 18-month deadline; commodity BSS/OSS capability (product catalog engines, order orchestration) is well served by mature vendor platforms, and building it in-house is pure schedule and cost risk with no competitive upside.

Implications: Vendor evaluation (Phase E) is a first-class activity, not an afterthought. Build-vs-buy is documented per solution building block. In-house builds require ARB justification against this principle.

P-06 — Security and Data Privacy Are Designed In, Not Bolted On

Statement: Every new component is designed to enforce data privacy (subscriber PII, location data implied by network-slice usage) and role-based access control from first release, not retrofitted after launch.

Rationale: Telecom subscriber and location data carries regulatory obligations (telecom-sector privacy regulation and general data protection obligations); retrofitting access control after a system holds live subscriber data is materially more expensive and carries breach exposure in the interim.

Implications: Every architecture contract (Phase G) requires a completed data classification and access control design before implementation approval. Network-slice telemetry, which can reveal subscriber location and behavior, is classified as sensitive PII by default.

P-07 — One Product Catalog, Multiple Channels

Statement: There is exactly one authoritative product catalog capable of describing any sellable product — mobile, broadband, or 5G network slice, individually or bundled — and every sales channel (retail, digital, partner) consumes it via the same API contract.

Rationale: This is the direct architectural answer to Brenmark's core business problem: converged bundles are impossible while three systems each hold an incompatible partial view of "product."

Implications: The legacy billing system's internal product tables become a consumer, not a source, of catalog data during the transition (see ADR-002 on system-of-record placement). Any channel-specific product logic must be expressed as catalog configuration, not channel-side code.

P-08 — Network Slicing Is Treated as a Sellable Product from Day One

Statement: The product catalog and order management data models are designed to represent a 5G network slice as a first-class orderable, billable product — with its own SLA parameters, capacity, and pricing dimensions — not bolted on as a special case after launch.

Rationale: The board's 18-month mandate is specifically 5G-slice monetization; if the to-be data model cannot natively express a network slice as a product, the program fails its primary mandate regardless of what else it delivers.

Implications: Data architecture (Phase C) must validate the product and order models against network-slicing use cases before ARB sign-off, not as a later extension. Network orchestration integration (see ADR-005) is scoped into Phase D, not deferred to a "phase 2."

P-09 — Automate Provisioning; Eliminate Manual Handoffs as the Default Path

Statement: The target architecture automates end-to-end order-to-activation for standard product configurations. Manual handoffs are the explicit exception path for non-standard orders, not the default for all orders.

Rationale: Days-long manual provisioning is a named business problem in this program's mandate; an architecture that reduces provisioning time only for some products, while leaving the default path manual, does not solve the problem it was funded to solve.

Implications: Order orchestration (Phase C/E) must support closed-loop automation to network inventory and activation systems. Exception-path volume is tracked as an operational KPI (Phase H) and is expected to shrink, not stay flat, release over release.

P-10 — Every Integration Has One Owner and One Governance Record

Statement: Every system-to-system integration in scope has a named accountable owner and is registered in the architecture's integration inventory; no integration exists that the EA function cannot enumerate.

Rationale: Brenmark's current-state integration sprawl (undocumented point-to-point links between billing, CRM, and order management) is itself part of why change is currently slow and risky; an unmanaged integration inventory would simply reproduce the current problem inside the new architecture.

Implications: The API gateway (Phase D) is also the enforcement point for this principle — unregistered point-to-point traffic is treated as a non-compliance finding under Phase G governance.

P-11 — Non-Functional Requirements Are Binding, Not Aspirational

Statement: Availability, latency, and throughput targets for BSS/OSS components (e.g., real-time charging availability, order API latency) are stated as binding acceptance criteria in every architecture contract, with explicit measurement methods.

Rationale: BSS/OSS systems are revenue-critical infrastructure; an architecture that meets functional scope but silently degrades availability or performance is a regression Brenmark's customers and board will not tolerate.

Implications: Solution building blocks (Phase E) are evaluated against stated NFR targets, not just functional fit. Architecture contracts (Phase G) are non-compliant by default if NFR acceptance testing is skipped.

P-12 — Change Is a Planned Program Activity, Not a Side Effect of Delivery

Statement: Organizational change (new processes, retraining, role changes for CRM/billing/network operations staff) is planned and resourced as explicitly as the technical migration, with its own roadmap and success metrics.

Rationale: A converged BSS/OSS platform changes how customer service, billing operations, and network operations staff do their jobs; treating this as an unplanned by-product of the technical rollout is a proven cause of adoption failure in platform modernization programs.

Implications: Phase H is resourced and tracked with the same rigor as the technical phases, with its own budget line in the business case (Phase A).

2.2 Governance Framework Setup

The Architecture Review Board (ARB) is chaired by the Chief Enterprise Architect and has six voting members: the Chief EA (chair, tie-break vote), the VP Billing & Revenue Systems, the VP Customer Operations, the VP Network Engineering, the Head of InfoSec, and the Head of Data & Analytics. Quorum is 5 of 6 voting members. Three non-voting advisory seats — the Program Director for BSS/OSS Modernization, a Procurement/Vendor Management lead, and rotating solution architects — attend to inform decisions without a vote. A board-level Technology Committee sits above the ARB for rare escalations that change capital commitments or the 18-month deadline itself, and the CTO sits between the two: accountable for operational matters but delegating day-to-day architecture judgment to the ARB.

The ARB operates on four cadences: a standing bi-weekly, 90-minute session requiring quorum; a 3-business-day written exception ballot when a session slot is blocked; four mandatory compliance checkpoint reviews per solution building block (contract sign-off, design conformance, pre-production readiness, post-go-live health); and a quarterly full program health review reported to the CTO and Board. Exceptions to any approved standard follow a defined process: a solution architect submits a written request citing the standard, the reason compliance isn't the right call, the proposed alternative, and a proposed expiry date; the request is reviewed at the next standing session or via written ballot if the team is blocked; approval requires simple majority of voting members, except that exceptions touching P-01 (data ownership) or P-06 (security/privacy) require unanimous voting-member approval; every approved exception carries a mandatory expiry date, and an exception with no renewal request logged before expiry automatically becomes a Blocking non-compliance finding rather than silently continuing.

Non-compliance is calibrated into three severity tiers so governance responds proportionally. Advisory findings are low risk, logged, with a 60-day remediation window that does not block go-live. Must-Fix-Before-Production findings block the Pre-Production Readiness Review from passing until remediated and re-presented. Blocking findings — a P-01/P-06 violation without an approved exception, or an unregistered P-10 integration — halt deployment immediately. Disagreement escalates first from solution-architect level to the ARB (on principle conflict or cross-boundary impact), and from the ARB to the CTO only when the ARB itself cannot reach a majority; the rare further escalation to the Board Technology Committee is reserved for capital-commitment or deadline-changing decisions.

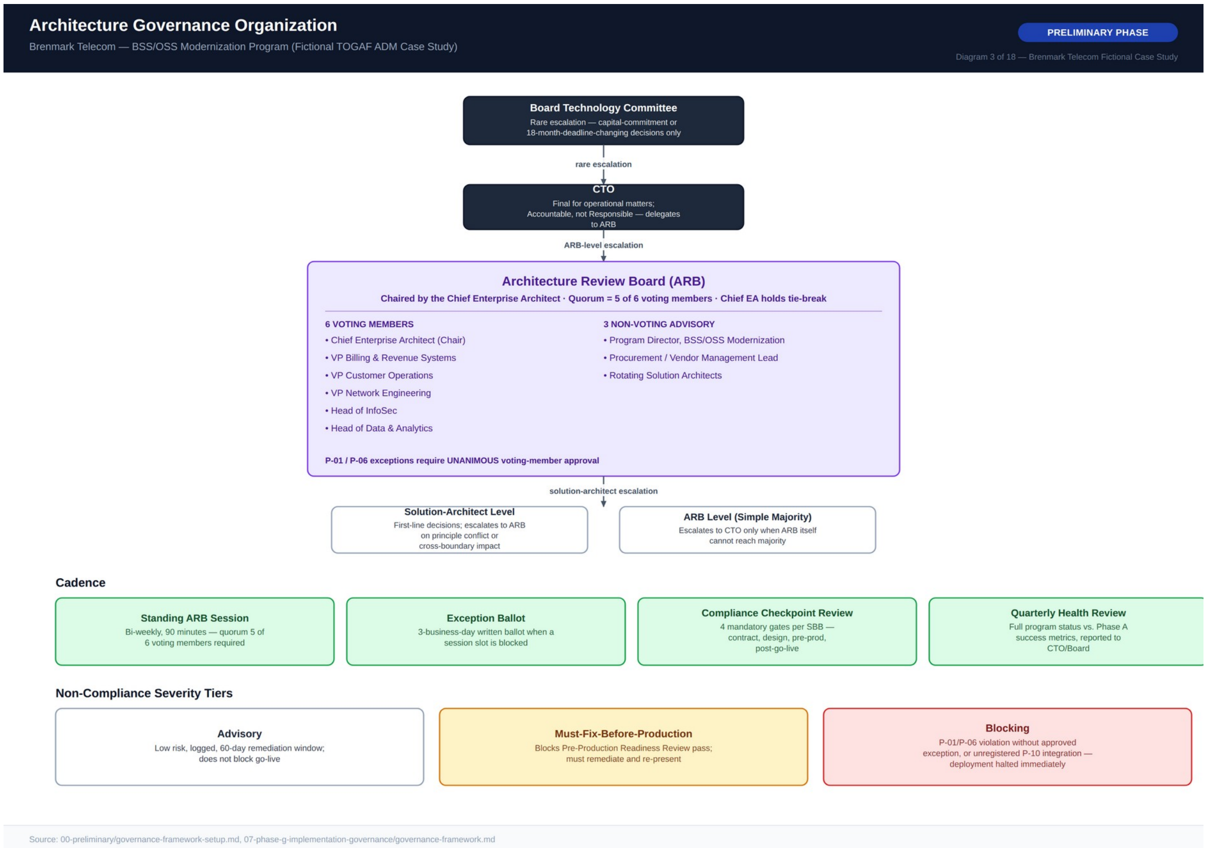


Diagram 3 — Architecture Governance Organization

3. Phase A — Architecture Vision & Scope

3.1 Problem Statement & Target-State Vision

Brenmark Telecom serves approximately 6 million subscribers across mobile and fixed broadband, on a BSS/OSS estate that grew organically over roughly two decades: a monolithic billing system as the financial system of record, a CRM system that maintains its own separate customer record, and a network inventory/order management system that requires manual, department-to-department handoffs to provision service. None of the three shares a common product catalog or a common customer/order data model, producing three consequences each mapped to a commercial deadline: no converged bundles are possible, since no system can express a product spanning mobile, broadband, and slice simultaneously, while competitors offering converged bundles win share in Brenmark's highest-growth segments; standard provisioning takes 3–7 business days of manual cross-system coordination, both a customer-experience liability and a direct labor cost; and there is no path to 5G-slice monetization at all, against a board mandate to deliver it within 18 months.

Within 18 months, Brenmark will operate a BSS/OSS architecture with a single, authoritative, TM Forum Open API–aligned product catalog describing any sellable product — mobile, broadband, or 5G network slice, individually or bundled — consumed identically by every sales channel; order-to-activation for standard configurations automated end-to-end, cutting the default provisioning path from days to minutes, with manual handoff retained deliberately as the exception path rather than eliminated outright; 5G network slices orderable, billable, and orchestrated from the same catalog and order-management flow as any other product; and the legacy billing system remaining the financial system of record, reached via a governed API layer, without a disruptive cutover. This vision is deliberately an evolution of the existing billing estate, not a replacement of it (ADR-001).

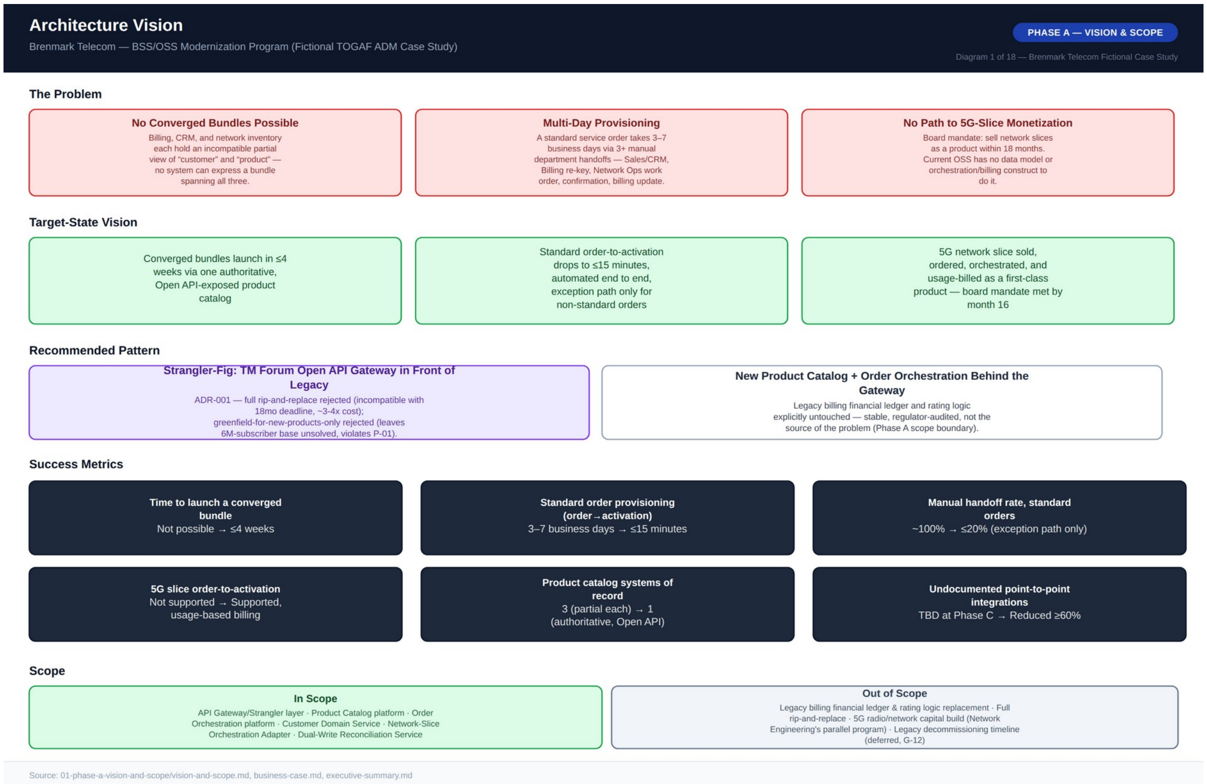


Diagram 1 — Architecture Vision One-Pager

3.2 Scope Boundaries

In scope: product catalog consolidation into a single TM Forum Open API–aligned system of record; order management and provisioning orchestration redesign for automated order-to-activation; customer/order/product data architecture (entity models, systems of record, integration patterns); a TM Forum Open API gateway/strangler layer in front of the legacy billing, CRM, and network inventory systems; network-slice orchestration integration sufficient to order, activate, and usage-bill a 5G slice; vendor selection for the new product catalog and order orchestration layer; and organizational change management for CRM, billing operations, and network operations staff.

Explicitly out of scope: replacing the core billing engine's financial ledger/rating logic — stable, regulator-audited, and not the source of the bundling/provisioning problem, so replacing it adds multi-year risk for no benefit to the 18-month mandate (the central premise of ADR-001); a retail/physical point-of-sale hardware refresh, which has its own separate business case; the network core/RAN 5G build-out itself, a parallel Network Engineering capital program this architecture integrates with but does not design or fund; CRM replacement, since only its lack of a shared customer/order view is in scope, closed via the shared data architecture rather than a CRM swap; an international/roaming billing overhaul, flagged as a candidate for a follow-on program; and full decommissioning of the legacy order-management system, a multi-year outcome this program enables but does not commit to completing within 18 months.

3.3 Success Metrics

Metric	Baseline (As-Is)	Target (18 Months)
Time to launch a new converged bundle product	Not currently possible	≤ 4 weeks, definition to market availability
Standard order provisioning time (order to activation)	3–7 business days	≤ 15 minutes for standard configurations
Manual handoff rate for standard orders	~100%	≤ 20% (exception path only)
5G network slice: order-to-activation capability	Not supported	Supported, with usage-based billing
Product catalog systems of record	3 (billing, CRM, order mgmt, each partial)	1 (authoritative, Open API–exposed)
Point-to-point undocumented integrations	Baseline TBD at Phase C completion	Reduced ≥ 60% vs. Phase C baseline

3.4 Stakeholder Map

Fourteen stakeholder groups were assessed, from the CEO/Board Technology Committee (Accountable for meeting the 18-month mandate and program budget) and the CTO (Accountable for overall program risk and technical feasibility, delegating execution to the ARB) through to frontline CRM/Billing/Network Ops staff and external customers, represented through VP Customer Operations and the Head of Product/Commercial respectively. A full RACI matrix governs program-level decisions: the CTO is Accountable but not Responsible for most program-level decisions, delegating to the ARB and Program Director; the VP Billing & Revenue Systems is Responsible for every wave's go-live decision, regardless of ARB approval, because billing carries the highest business-continuity risk of any component in scope; and the Head of InfoSec holds an explicit consulted-with-veto role on any P-01 or P-06 exception, reflecting that those two principles require unanimous ARB approval to override.

The VP Billing & Revenue Systems' primary concern is zero disruption to real-time charging, regulatory audit continuity, and billing remaining the system of record for financial data; the VP Network Engineering is focused on network inventory accuracy and orchestration integration not destabilizing existing provisioning; and the Head of Data & Analytics is accountable for data quality and governance of the new authoritative product/customer/order domains, explicitly tasked with avoiding a new generation of shadow copies as capability migrates off legacy systems.

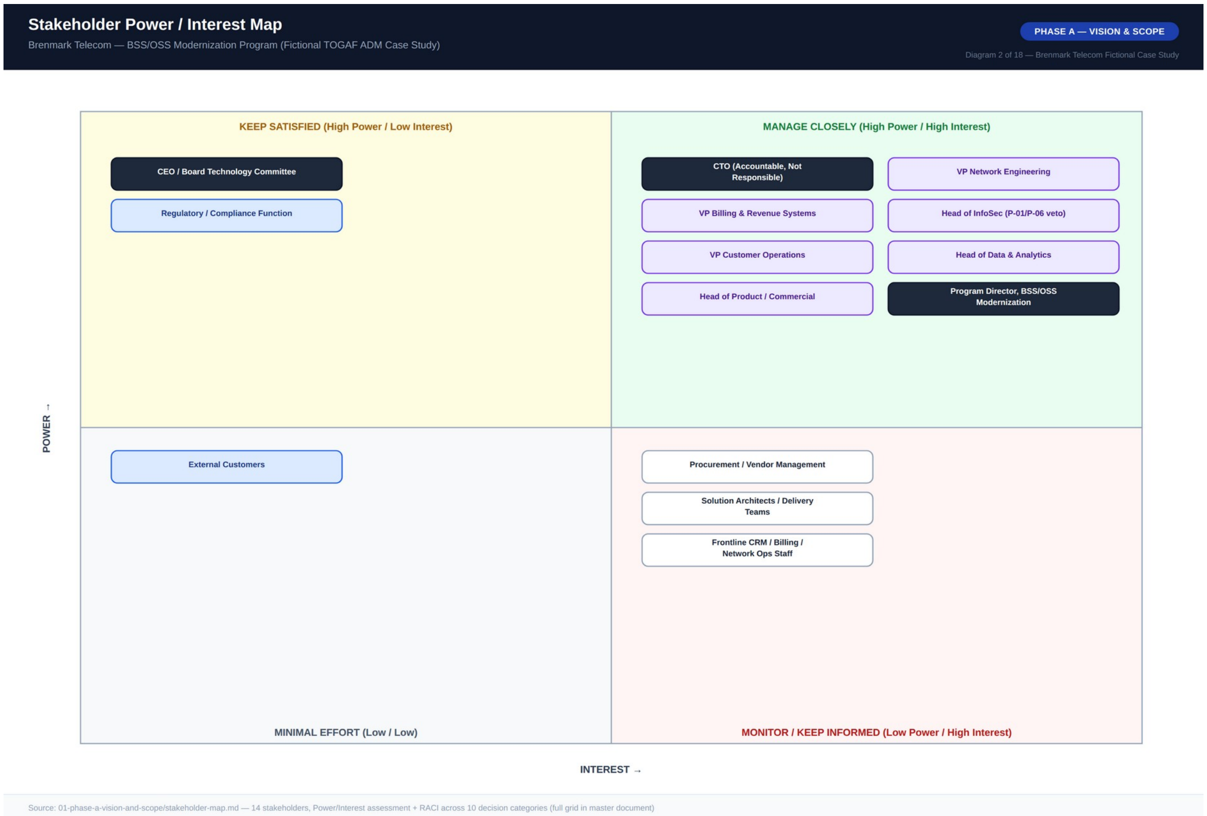


Diagram 2 — Stakeholder Power / Interest Map

3.5 Business Case Summary

The program requires an estimated $18.4M in capital investment over 18 months, dominated by the $4.6M API gateway/strangler layer build and the combined $7.0M license-plus-implementation cost of the Product Catalog and Order Orchestration platforms, plus an estimated 15% ongoing operational overhead on affected systems during a ~24-month dual-write transition period (ADR-004). Against this, the program projects $50.7M in combined benefit over a 3-year horizon: $27.4M in lower total cost of ownership versus doing nothing ($53.4M as-is run cost over 3 years versus $26.0M to-be total cost, already inclusive of capex and transition opex) plus $23.3M in new converged-bundle and 5G-slice revenue. The largest single benefit driver is new revenue, not cost savings — bundle uptake is projected to reach 25% of the base by end of Year 3 with a $6/month ARPU uplift per bundled subscriber, while enterprise/IoT slice accounts ramp to 120 by end of Year 3 at an average $18K/account/year.

Payback is projected at approximately 22 months from program start: a cumulative $13.7M net outflow at end of Year 1 narrows to $10.3M cumulative net outflow at end of Year 2 as new revenue and legacy-saving deltas build, crossing breakeven roughly two months after the board's 18-month capability deadline — a deliberate consequence of the mandate being a capability deadline (slice monetization must be possible by month 18), not a breakeven deadline. Any capex variance exceeding 10% of the approved $18.4M triggers mandatory escalation to the Board Technology Committee.

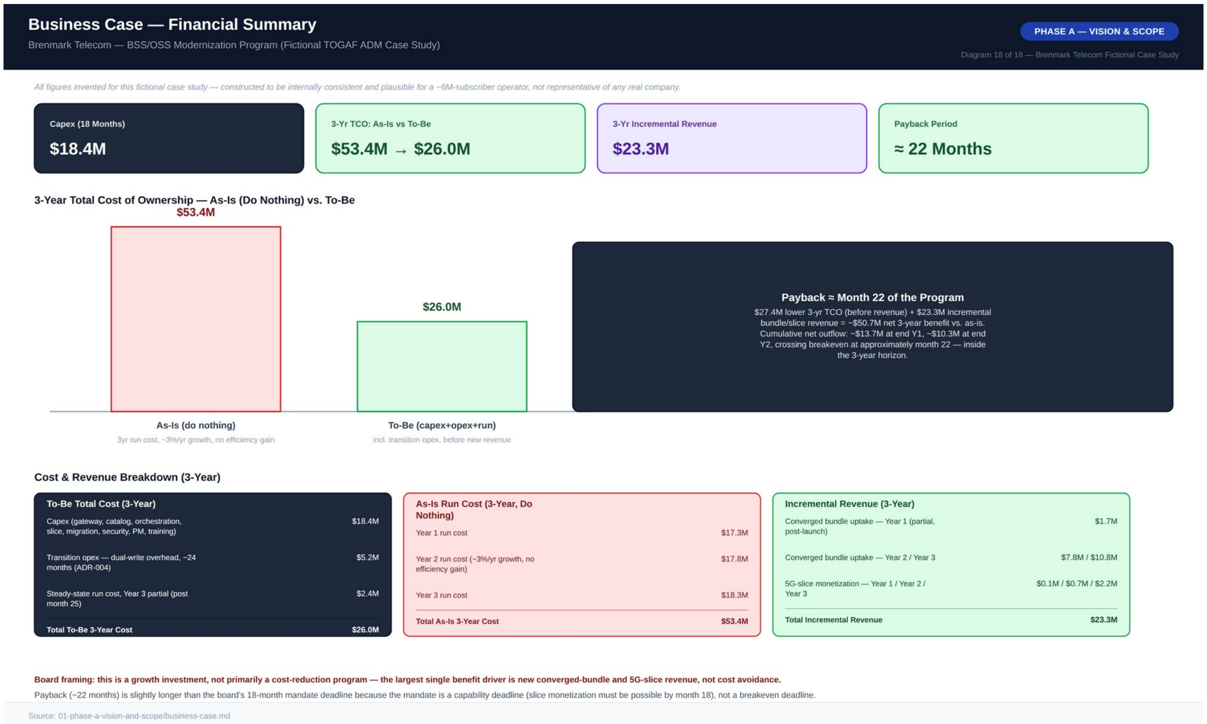


Diagram 18 — Business Case Financial Summary

4. Phase B — Business Architecture

4.1 As-Is Business Architecture

Brenmark's order-to-activation process today runs entirely on manual cross-system coordination. Sales/CRM captures an order against CRM's own product code list, which is not shared with Billing or Network Inventory; a customer service representative then manually re-keys the order into the billing system, translating between code lists via undocumented institutional knowledge; a manual work order goes to Network Operations, where Network Inventory checks capacity in a system not connected to either CRM or Billing; Network Ops manually confirms completion via email or ticket, relayed to Customer Operations, which finally updates Billing so charging can begin. This takes 3–7 business days end-to-end for any product requiring network provisioning, and no converged bundle has ever shipped, because the three systems share no common product definition. There is no shared customer ID across CRM and Billing — reconciliation happens only by manual phone or account-number lookup — and the undocumented product-code translation step depends on tenured staff, a retirement risk the architecture function has flagged explicitly.

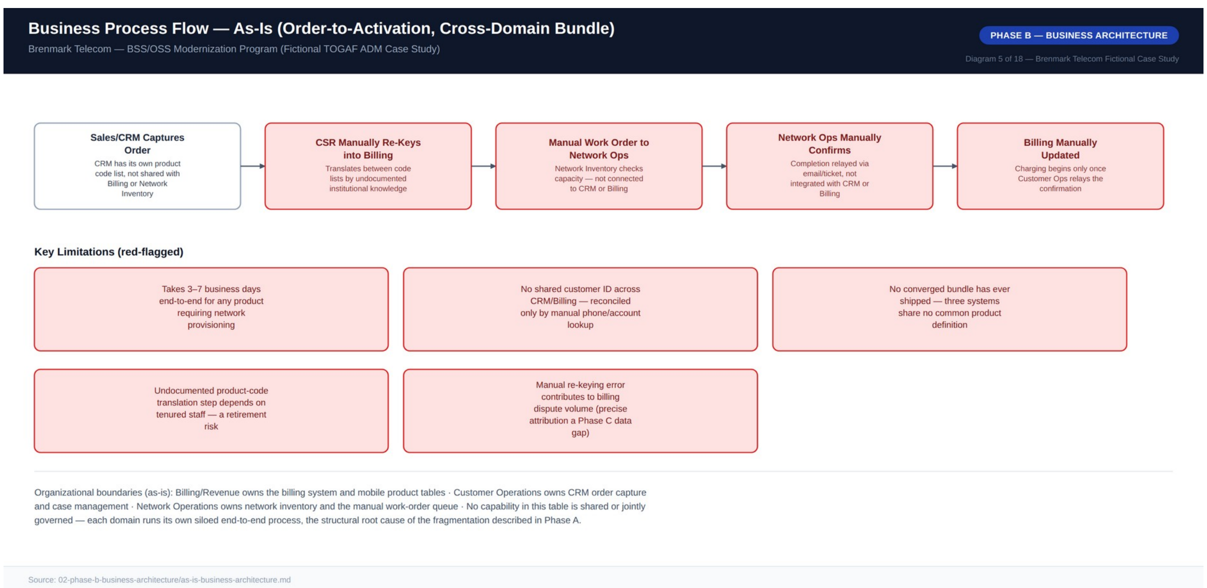


Diagram 5 — Business Process Flow, As-Is (Order-to-Activation, Cross-Domain Bundle)

4.2 To-Be Business Architecture

In the target state, an order is captured directly against the single Product Catalog — one consistent product definition, whether mobile, broadband, slice, or bundle, across every channel — and the Order Orchestration Platform decomposes it into billing, network, and slice actions issued as API calls, not manual tickets. Billing receives a charge instruction via API and begins financial account setup without re-keying; standard network and slice activation complete automatically via API, with no human work order; and orchestration confirms completion via API callback from every domain, not email, with the manual exception path retained only for non-standard orders. Standard order-to-activation drops from 3–7 business days to a target of ≤15 minutes, and converged bundles become possible for the first time — one order, one orchestration flow. What deliberately stays the same: the financial ledger is untouched, and Billing keeps its system of record, reached only via API, consistent with the capability map's decision to hold Billing & Charging maturity flat by design.

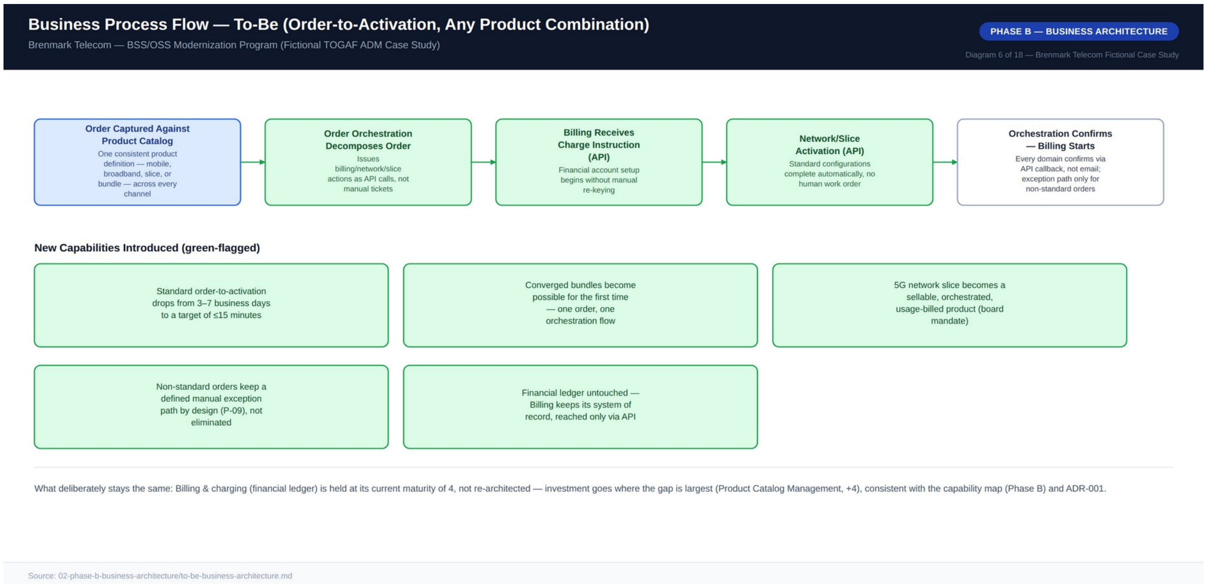
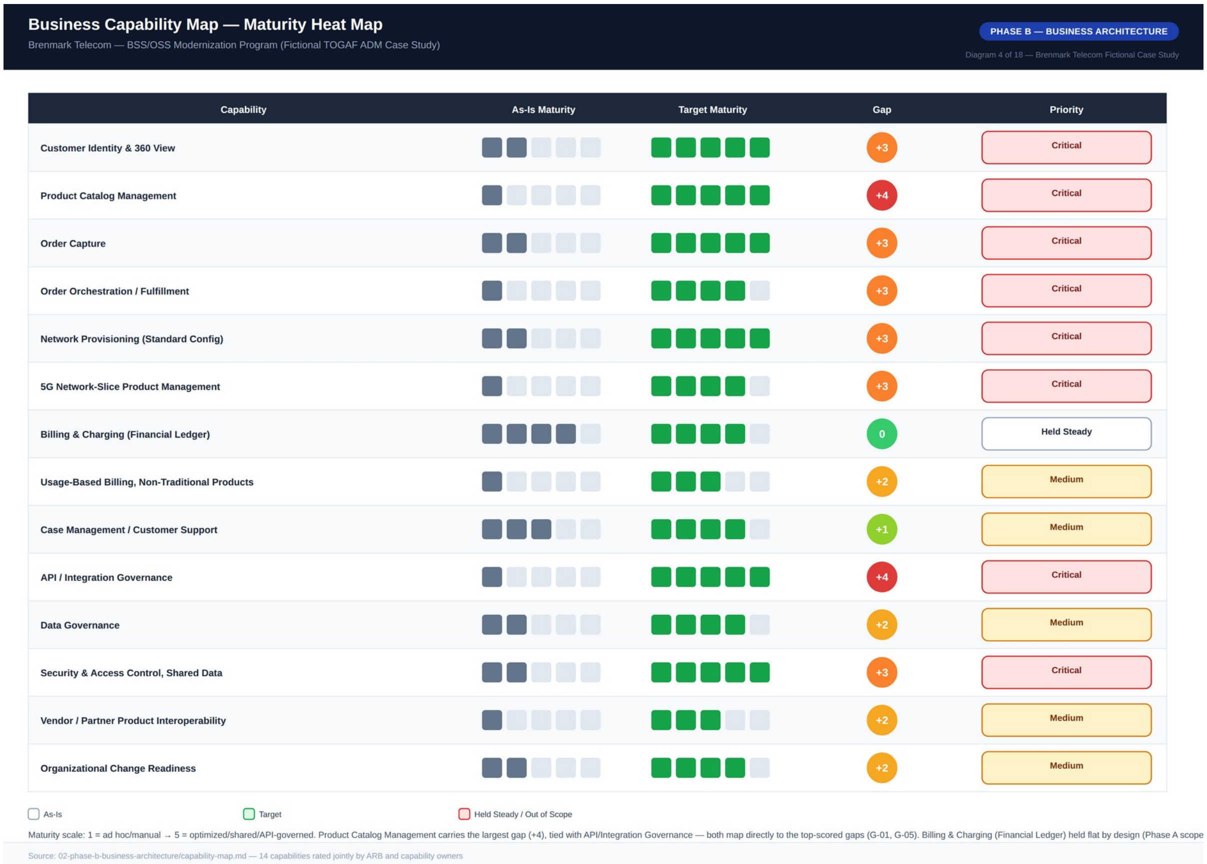


Diagram 6 — Business Process Flow, To-Be (Order-to-Activation, Any Product Combination)

4.3 Business Capability Map

Fourteen business capabilities were rated jointly by the ARB and capability owners on a 1–5 maturity scale (1 = ad hoc/manual through 5 = optimized, shared, and API-governed). Product Catalog Management carries the single largest gap (1→5, +4) — today product definition exists only as disconnected tables in two systems with no catalog concept at all — tied with API/Integration Governance (also +4, since no integration inventory or governance exists today). This is why Phase E treats catalog platform selection as the highest-stakes single decision in the program (ADR-002). Billing & Charging (Financial Ledger) is intentionally left at its current maturity of 4, deliberately unchanged and out of scope, consistent with ADR-001's strategy of not needing to touch a well-run capability to succeed. Order Orchestration and 5G Network-Slice Product Management are deliberately targeted at 4, not 5 — both retain a defined manual exception path or defer full optimization to a follow-on program, consistent with the incremental modernization principle (P-03) and P-09's explicit target of ≤20% manual handling, not zero.



Capability	As-Is Maturity	Target Maturity	Gap	Priority
Customer Identity & 360 View	2	5	+3	Critical
Product Catalog Management	1	5	+4	Critical
Order Capture	2	5	+3	Critical
Order Orchestration / Fulfillment	1	4	+3	Critical
Network Provisioning (Standard Config)	2	5	+3	Critical
5G Network-Slice Product Management	1	4	+3	Critical
Billing & Charging (Financial Ledger)	4	4	0	Held Steady
Usage-Based Billing, Non-Traditional Products	1	3	+2	Medium
Case Management / Customer Support	3	4	+1	Medium
API / Integration Governance	1	5	+4	Critical
Data Governance	2	4	+2	Medium
Security & Access Control, Shared Data	2	5	+3	Critical
Vendor / Partner Product Interoperability	1	3	+2	Medium
Organizational Change Readiness	2	4	+2	Medium

Diagram 4 — Business Capability Map, Maturity Heat Map

4.4 Business Footprint

Customer Operations owns sales/digital channel order capture and case management, consuming the new CRM and Order Orchestration Platform against an Enterprise CRM plus TM Forum Open API gateway access. Billing & Revenue Systems retains real-time charging and financial ledger ownership, unchanged except for API-fronting via the Legacy Billing Adapter. Network Engineering owns standard-config automated activation and 5G-slice instantiation through the Network Inventory System and the new Network-Slice Orchestration Adapter. A new Shared/Jointly-Governed organizational unit owns the Product & Order capability spanning the Product Catalog Platform, Order Orchestration Platform, and Customer Domain Service — realized on Northolt BSS Cloud (Vendor B) behind the API Gateway/Strangler Layer. Security & InfoSec owns the RBAC uplift and PII classification enforcement across the shared data layer (P-06), and Architecture & Governance owns ARB review, architecture contracts, the exceptions register, and compliance checkpoints via governance tooling and the integration inventory.

Business Footprint Diagram

Brenmark Telecom — BSS/OSS Modernization Program (Fictional TOGAF ADM Case Study)

PHASE B — BUSINESS ARCHITECTURE

Diagram 7 of 18 — Brenmark Telecom Fictional Case Study

Organization Unit	Business Function	Key Process / Service	Supporting Application (To-Be)	Technology Platform
Customer Operations	**Customer Care & Order Capture**	Sales/digital channel order capture, case management via Customer 360	CRM (consumer), Order Orchestration Platform	Enterprise CRM + TM Forum Open API gateway access
Billing & Revenue Systems	**Financial Ledger & Charging**	Real-time charging; billing account setup via API instruction, no re-keying	Billing System (retained, API-fronted)	Legacy billing engine + Legacy Billing Adapter
Network Engineering	**Network Provisioning & Activation**	Standard-config automated activation; 5G-slice instantiation	Network Inventory System + Network-Slice Orchestration Adapter	Network inventory platform, 5G core orchestration
Shared / Jointly-Governed	**Product & Order Capability**	Single product definition across mobile/broadband/slice; cross-domain order orchestration	Product Catalog Platform, Order Orchestration Platform, Customer Domain Service	Northolt BSS Cloud (Vendor B) + API Gateway/Strangler Layer
Security & InfoSec	**Access Control & Classification**	RBAC uplift across the shared data layer; PII classification enforcement (P-06)	Centralized RBAC provider, data classification tooling	Enterprise identity provider (extended)
Architecture & Governance	**Program Governance**	ARB review, architecture contracts, exceptions register, compliance checkpoints	Governance tooling, integration inventory, exceptions register	Enterprise collaboration platform

Domain Owner (Retained) · Unchanged / Legacy · New Shared Capability · Security / Compliance · Governance

Source: 02-phase-b-business-architecture/to-be-business-architecture.md, capability-map.md, 03-phase-c/application-architecture.md

Diagram 7 — Business Footprint Diagram

5. Phase C — Information Systems Architecture

5.1 Application Architecture

The as-is landscape, shown in Diagram 8, consists of three large, independently operated applications with no API gateway, no service catalog, and no consistent authentication model: the Billing System (monolith) holds the financial ledger, real-time charging, and mobile product tables, integrating to CRM only via nightly batch file exchange; the CRM System holds its own customer record and syncs to Billing the same way, with no integration at all to Network Inventory beyond manual data entry; and the Network Inventory/Order Management System tracks network resources and broadband service definitions with work orders raised and closed manually via ticketing. There is no TM Forum ODA alignment and no distinct "Product Catalog" or "Order Management" component — catalog and order logic sit embedded as monolithic internal modules inside Billing and CRM respectively.

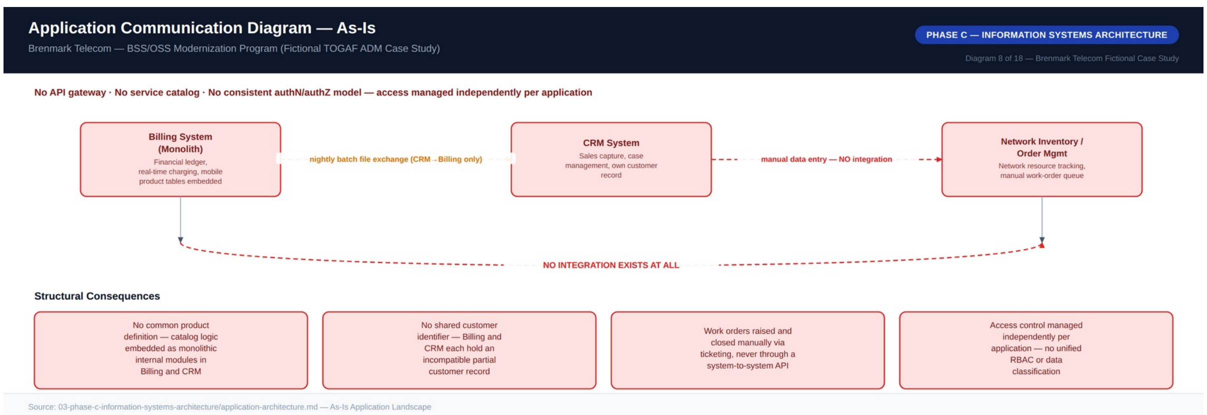


Diagram 8 — Application Communication Diagram, As-Is

The to-be application architecture introduces three new ODA-aligned components — the Product Catalog Platform (Core Commerce Management), the Order Orchestration Platform (Production/Order Management), and the Customer Domain Service (Party Management) — connected to the retained legacy applications exclusively through a governed API Gateway/Strangler Layer rather than new point-to-point links (ADR-003). The Product Catalog platform never writes directly into the Billing system's product tables; it publishes catalog data through the gateway, and a scheduled reconciliation job keeps the legacy tables consistent during the transition (ADR-004). This means legacy internal complexity is never a blocker to building new capability, and legacy capability can eventually be retired behind the gateway's stable API contract without new components needing to change when that retirement happens. Replacing the Billing system's financial ledger wholesale was seriously evaluated and rejected as a multi-year, high-risk undertaking incompatible with the 18-month deadline (ADR-001); the API Gateway/Strangler pattern instead lets Brenmark ship new capability on its own schedule, decoupled from legacy release cycles.

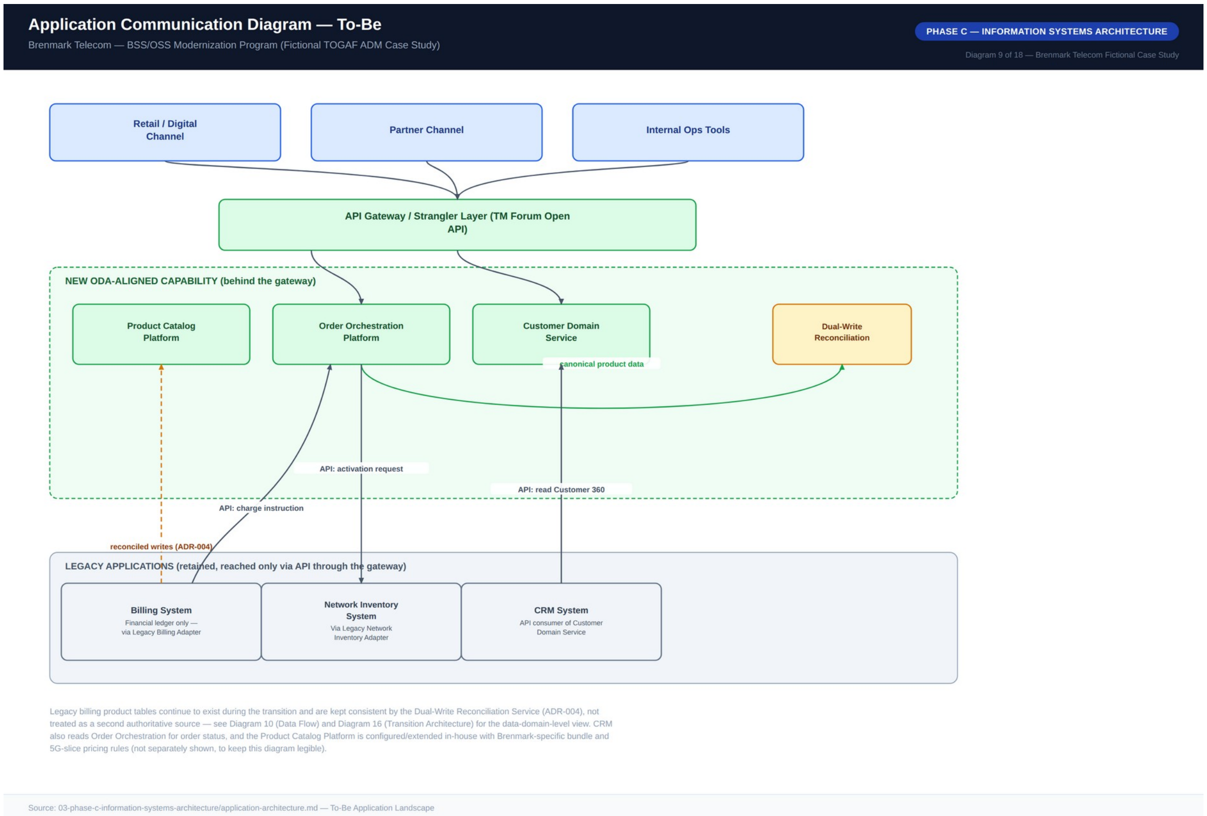


Diagram 9 — Application Communication Diagram, To-Be

5.2 Data Architecture

The core data-architecture problem is that Customer, Product, and Order are each represented as incompatible, disconnected data models today: Customer exists as two separate records (CRM's sales/case record and Billing's financial account record) reconciled only by manual lookup, with no shared unique customer ID; Product spans Billing's mobile-focused product/pricing tables and Network Inventory's broadband service definitions with no concept spanning both, so a bundle cannot be expressed at all; and Order state is fragmented across CRM order capture, manual network work-order tickets, and Billing's activation record, with no single order-lifecycle record and status cross-checked manually.

The to-be target establishes five canonical data domains, each with exactly one system of record per P-01: Customer (new Customer Domain service, single authoritative record with a unique global customer ID); Product Catalog (new Product Catalog platform, TM Forum Open API–aligned, TMF620/TMF622-pattern); Order (new Order Orchestration platform, single order-lifecycle record, TMF622-pattern); Network Inventory (existing system, retained and exposed via API rather than replaced); and Financial Ledger (existing Billing engine, retained and unchanged per Phase A scope). Data moves via a hybrid pattern: synchronous TM Forum Open API–styled REST calls for order capture, catalog lookups, and activation requests where an immediate response is needed, and asynchronous events for state-change propagation (e.g., "order activated," "billing account created") that decouple domain systems from each other's response times. Because the legacy billing system's own product tables cannot be immediately retired, the Product Catalog platform is authoritative and the legacy tables are updated by a scheduled reconciliation job rather than becoming a second authoritative source — the specific mechanism, consistency model, and retirement timeline governed by ADR-004.

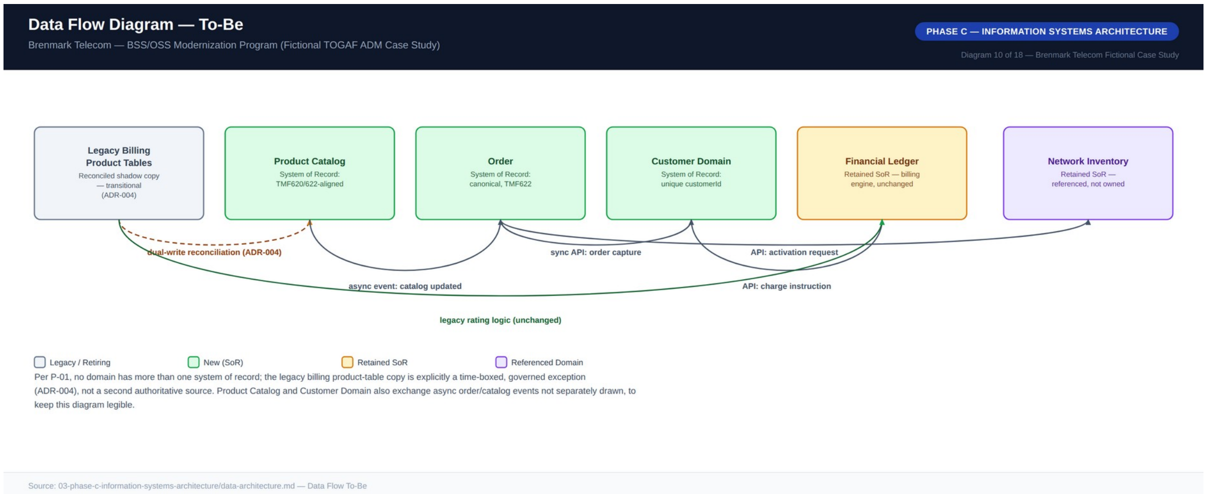


Diagram 10 — Data Flow Diagram, To-Be

5.3 Data Ownership / Domains

Per P-01, no domain may have more than one system of record; every other system consuming that domain's data does so through a governed API, not a private copy. The Customer Domain Service owns the global customer identifier, legal name, billing addresses, and linked account references. The Product Catalog Platform owns product type (mobile, broadband, 5G-slice, or bundle), pricing reference, SLA parameters — critically including slice capacity/latency/throughput parameters per P-08 — and bundle composition. The Order Orchestration Platform owns the order-lifecycle state machine (captured → decomposed → in-fulfillment → active → billed) with per-domain fulfillment sub-status. Network Inventory and the Financial Ledger remain owned by their existing retained systems, reached via API for capacity/activation checks and charge instructions respectively. The Head of Data & Analytics is accountable for data quality across the new shared domains, and every new consumer of Customer, Product, or Order data must be registered in the integration inventory (P-10) and access-controlled per P-06 — network-slice usage telemetry is classified sensitive PII by default and requires explicit InfoSec sign-off before a new consumer is onboarded.

Diagram 11 — Data Ownership / Domain Diagram

6. Phase D — Technology Architecture

6.1 Reference Architecture Pattern

The core technology pattern places a TM Forum Open API–conformant gateway between Brenmark's channels/new components and its legacy BSS/OSS estate, with legacy systems "strangled" incrementally rather than replaced in a single cutover (ADR-001). This pattern applies when the legacy system is revenue-critical and cannot tolerate a big-bang cutover; a hard external deadline rules out a multi-year replacement; the new capability needed is additive to, not a full replacement of, existing legacy function; the organization can tolerate a bounded period of dual-write/reconciliation complexity in exchange for avoiding cutover risk; and a standards body already defines the target API contracts — all five conditions held true for Brenmark at Phase A.

The ARB explicitly documented when this pattern is the wrong choice, to prevent it being applied where it doesn't fit: when the legacy system's internal data model is too corrupted or undocumented to reconcile against reliably; when there is no hard deadline forcing incremental delivery and the organization can afford a multi-year replacement for a permanently simpler end state; when the legacy system is being sunset for unrelated commercial reasons anyway; when regulatory or contractual constraints require a single authoritative system with no intermediate dual-write state at all (confirmed not to apply to Brenmark's billing regulator per Phase A scope confirmation with Compliance); and when the organization lacks the governance discipline to actually retire legacy capability once strangled, which is why Phase G explicitly tracks legacy-traffic-share-through-gateway as a compliance metric, not just a delivery milestone. The gateway carries binding non-functional targets under P-11: 99.95% availability, ≤200ms P95 catalog-read latency, ≤800ms P95 order-submission latency, and 100% TM Forum Open API schema validation at ingress, with non-conformant traffic rejected rather than silently passed through.

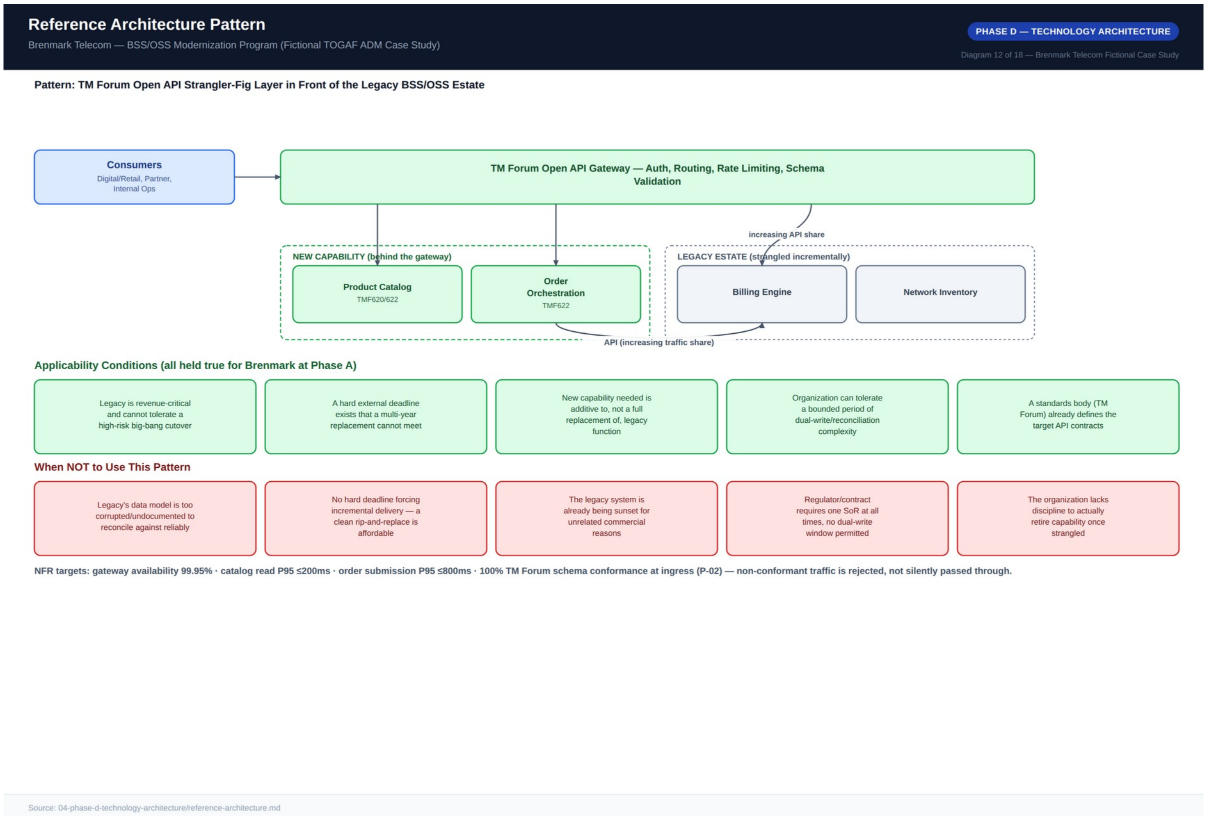


Diagram 12 — Reference Architecture Pattern

6.2 Technology Standards

The approved stack spans the TM Forum Open API family for integration contracts (P-02), REST/JSON over HTTPS for synchronous calls with event-driven pub/sub for asynchronous state propagation (P-02/P-10), a centralized gateway enforcing schema validation, auth, and rate limiting for all cross-system traffic (P-10), centralized RBAC with no application-local user/role store for new components (P-06), mandatory data classification (Public/Internal/Confidential/Restricted-PII) tagged at entity creation (P-06), the TM Forum Open Digital Architecture (ODA) functional decomposition for all new components (P-05/P-07), container-based deployment with full environment parity (P-11), centralized logging/tracing/metrics for every gateway-routed transaction (P-04/P-11), and mandatory OpenAPI/TM Forum schema documentation in the integration inventory for every registered integration (P-10). New point-to-point integrations bypassing the gateway, new application-local data stores duplicating a system of record, and batch file-based nightly integration patterns (the as-is CRM–Billing sync method) are all explicitly deprecated for new work.

Exceptions follow the same process described in Section 2.2, with a worked illustrative example: an exception was granted for the Network Inventory system’s activation-confirmation callback, which could only be delivered via a legacy SOAP interface because the vendor’s committed REST API delivery date fell outside the program’s first migration wave. The exception was approved by simple majority (not touching P-01/P-06), carries a 9-month expiry tied to the vendor’s committed REST delivery date, and required that the SOAP traffic still route through the API gateway — satisfying P-10 — even with a non-standard payload format. This illustrates the exceptions process working as intended: a real constraint, a bounded and justified deviation, and a concrete expiry rather than an indefinite carve-out.

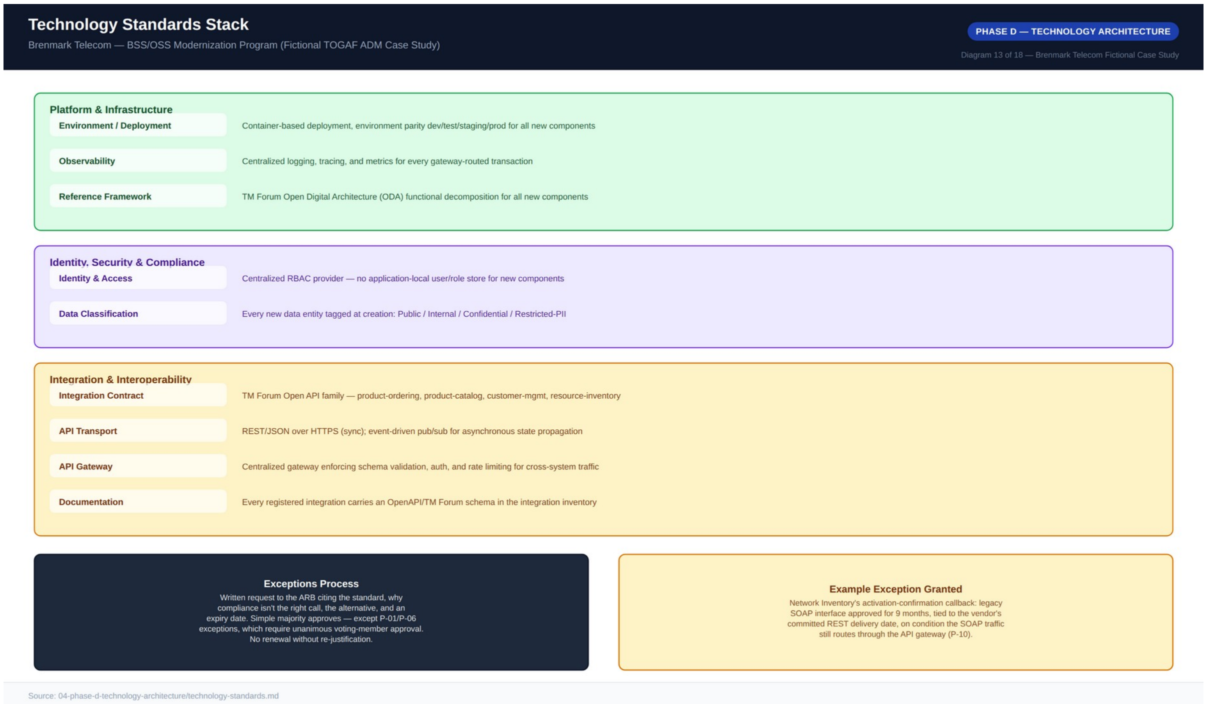


Diagram 13 — Technology Standards Stack

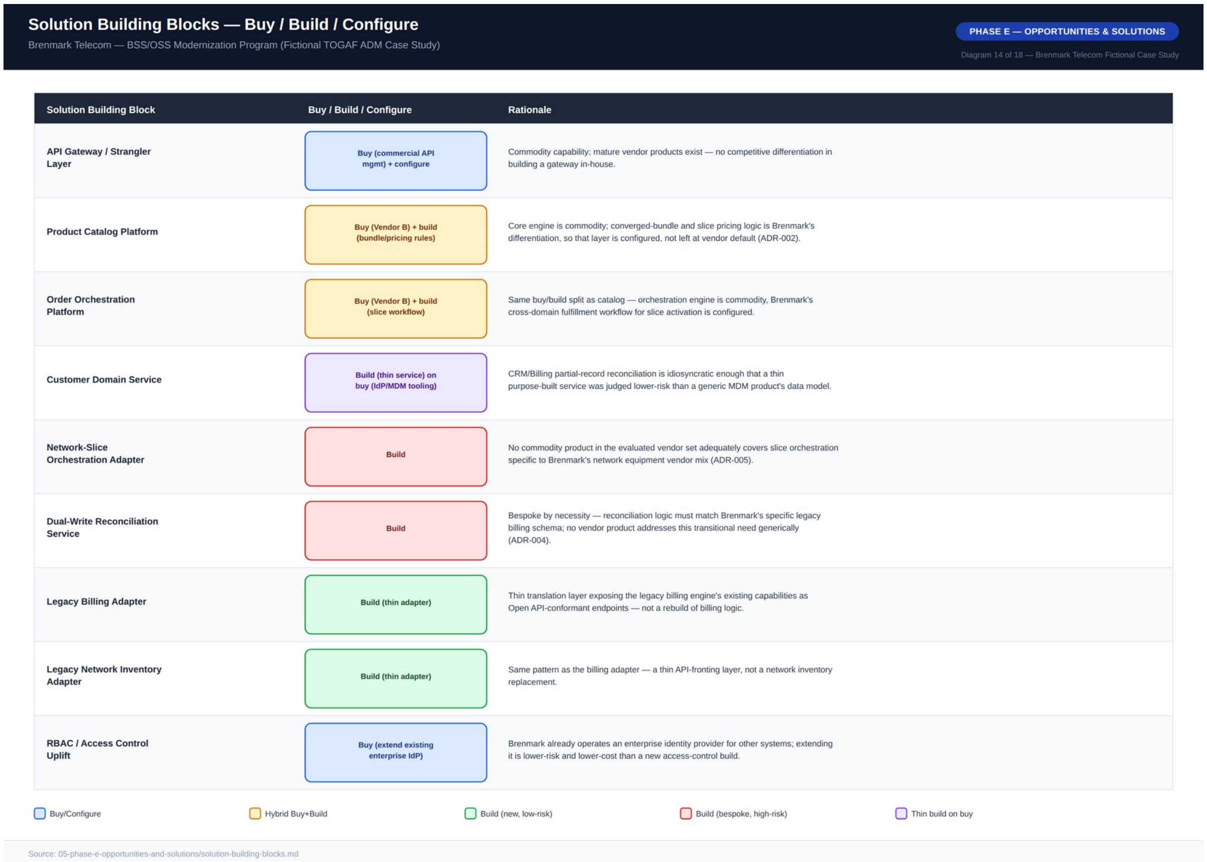
7. Phase E — Opportunities & Solutions

7.1 Gap Analysis

Twelve gaps were registered and scored on Impact (1–5) × Urgency (1–5) for a priority score out of 25. The top three gaps share the maximum score of 25 and map directly to Brenmark's three named business problems: no unified product catalog spanning mobile/broadband/slice (G-01, closed by the Product Catalog Platform SBB), no automated cross-domain order orchestration (G-02, closed by the Order Orchestration Platform SBB), and no 5G network-slice product/orchestration integration (G-03, closed by a build-in-house Network-Slice Orchestration Adapter) — all three addressed in the earliest waves the roadmap can technically absorb them. G-06 (legacy systems not API-addressable) scores lower individually (16) but is a hard technical prerequisite for G-01/G-02/G-03 to be testable at all, which is why it is scheduled in Wave 1 ahead of the higher-scored platform builds. G-12 (no defined legacy decommissioning path beyond this program) scores lowest (4) and is deliberately deferred as a candidate for a follow-on program, consistent with the Phase A scope boundary. A generic Master Data Management platform was considered for closing G-04 and rejected as adding 6+ months of vendor selection for a capability broader than Brenmark's two-source reconciliation problem actually requires; a separate "billing modernization" gap was raised in workshops but excluded from the register entirely as out of scope.

7.2 Solution Building Blocks

Each target capability decomposes into a discrete solution building block marked Buy or Build per P-05. The API Gateway/Strangler Layer is bought (commercial API management platform) and configured — a commodity capability with no competitive differentiation in building it in-house. The Product Catalog Platform and Order Orchestration Platform are both buy-plus-build: the core engines are commodity, but Brenmark-specific converged-bundle/slice pricing logic and cross-domain fulfillment workflow are where Brenmark's commercial differentiation actually lives, so that layer is configured and extended, not left as vendor default. The Customer Domain Service is a thin build on top of existing identity/master-data tooling, since Brenmark's specific CRM/Billing record-reconciliation logic was judged idiosyncratic enough that forcing it into a generic MDM product's data model was a worse fit than a purpose-built thin service. The Network-Slice Orchestration Adapter, the Dual-Write Reconciliation Service, and the Legacy Billing and Network Inventory Adapters are all builds — no commodity product in Brenmark's evaluated vendor set adequately covers slice orchestration for Brenmark's specific network equipment mix, and reconciliation logic must match Brenmark's specific legacy schema by necessity. RBAC/Access Control Uplift is bought by extending Brenmark's existing enterprise identity provider rather than building new access control. The Legacy Billing Adapter and Legacy Network Inventory Adapter are prerequisite building blocks for nearly every other SBB — the Product Catalog and Order Orchestration platforms cannot be meaningfully tested end-to-end until they can reach billing and network inventory through a stable adapter contract, which directly shapes Wave 1's sequencing ahead of the platforms that depend on it.



Solution Building Blocks — Buy / Build / Configure

Brenmark Telecom — BSS/OSS Modernization Program (Fictional TOGAF ADM Case Study)

PHASE E — OPPORTUNITIES & SOLUTIONS

Diagram 14 of 18 — Brenmark Telecom Fictional Case Study

Solution Building Block	Buy / Build / Configure	Rationale
API Gateway / Strangler Layer	Buy (commercial API mgmt) + configure	Commodity capability; mature vendor products exist — no competitive differentiation in building a gateway in-house.
Product Catalog Platform	Buy (Vendor B) + build (bundle/pricing rules)	Core engine is commodity; converged-bundle and slice pricing logic is Brenmark's differentiation, so that layer is configured, not left at vendor default (ADR-002).
Order Orchestration Platform	Buy (Vendor B) + build (slice workflow)	Same buy/build split as catalog — orchestration engine is commodity, Brenmark's cross-domain fulfillment workflow for slice activation is configured.
Customer Domain Service	Build (thin service) on buy (IdP/MDM tooling)	CRM/Billing partial-record reconciliation is idiosyncratic enough that a thin purpose-built service was judged lower-risk than a generic MDM product's data model.
Network-Slice Orchestration Adapter	Build	No commodity product in the evaluated vendor set adequately covers slice orchestration specific to Brenmark's network equipment vendor mix (ADR-005).
Dual-Write Reconciliation Service	Build	Bespoke by necessity — reconciliation logic must match Brenmark's specific legacy billing schema; no vendor product addresses this transitional need generically (ADR-004).
Legacy Billing Adapter	Build (thin adapter)	Thin translation layer exposing the legacy billing engine's existing capabilities as Open API-conformant endpoints — not a rebuild of billing logic.
Legacy Network Inventory Adapter	Build (thin adapter)	Same pattern as the billing adapter — a thin API-fronting layer, not a network inventory replacement.
RBAC / Access Control Uplift	Buy (extend existing enterprise IdP)	Brenmark already operates an enterprise identity provider for other systems; extending it is lower-risk and lower-cost than a new access-control build.

Diagram 14 — Solution Building Blocks, Buy / Build

7.3 Vendor Evaluation

Four vendors were evaluated for the combined Product Catalog and Order Orchestration Platform decision — the single highest-stakes buy decision in the program, since these two platforms make converged bundles and slice monetization technically possible at all. Northolt BSS Cloud ("Vendor B") won with a weighted score of 4.30/5.00, against Cascade Digital Commerce ("Vendor C," 3.50), Ferro OSS/BSS Platform ("Vendor D," 3.05), and Meridian Commerce Suite ("Vendor A," 3.00) — scoring highest on Open API conformance (20% weight), converged-bundle modeling (20% weight), and implementation timeline fit (15% weight), the three criteria most tied to program success, given its cloud-native, ODA-native architecture designed from the outset to sit behind an API gateway alongside legacy systems.

Vendor D (Ferro) scored highest of all four on 5G-slice orchestration specifically (5/5), reflecting genuine network-side orchestration strength, but scored weakest on converged-bundle commerce modeling (2/5) and Open API conformance (3/5) — the two highest-weighted criteria — making it a narrow fit for the slice mandate at the cost of the broader bundle problem (see ADR-005 for how Vendor D’s strength is preserved regardless via an in-house adapter). Vendor C (Cascade) scored competitively on TCO and timeline fit (4/5 each) with a strong catalog engine, but its order orchestration was a partner add-on rather than native capability, conflicting with treating orchestration as core rather than supplementary. Vendor A (Meridian) had the deepest telecom reference base and partner ecosystem (5/5 on both) but the weakest Open API conformance (3/5), worst implementation timeline fit (2/5, an estimated 16–20 months against an 18-month mandate), and worst lock-in/exit-cost score (2/5) of all four — disqualifying on timeline alone. This recommendation was presented to the ARB per the Phase A RACI (Program Director Responsible, ARB Chief EA Accountable, CTO Consulted); Vendor B's exit-cost score of 4/5 was a specific negotiating point to secure contractual data-portability commitments before signature, consistent with P-04.

Vendor Evaluation Scorecard — Product Catalog & Order Orchestration Platform

Brenmark Telecom — BSS/OSS Modernization Program (Fictional TOGAF ADM Case Study)

PHASE E — OPPORTUNITIES & SOLUTIONS

Diagram 17 of 18 — Brenmark Telecom Fictional Case Study

Weighted Criterion	Wt	Vendor A Meridian Commerce	Vendor B Northolt BSS Cloud	Vendor C Cascade Digital	Vendor D Ferro OSS/BSS
Open API conformance (out-of-box)	20%	3	5	4	3
Converged-bundle & multi-domain product modeling	20%	4	5	4	2
5G network-slice product/orchestration support	15%	2	4	2	5
Implementation timeline fit (≤12mo to go-live)	15%	2	4	4	3
Total cost of ownership (3-year)	10%	3	4	4	2
Vendor lock-in / exit cost (Principle P-04)	10%	2	4	3	3
Reference customer base in telecom BSS	5%	5	3	3	4
Partner / systems-integrator ecosystem depth	5%	5	3	3	3
Weighted Total		3.00	4.30 ★	3.50	3.05

Recommendation: Vendor B — Northolt BSS Cloud
Wins decisively on the three criteria most tied to program success: Open API conformance, converged-bundle modeling, and implementation timeline fit. Cloud-native, ODA-native architecture is a direct fit for the strangler-fig reference architecture (Phase D).

Closest Runner-Up: Vendor C — Cascade Digital Commerce
Competitive on TCO and timeline with a genuinely strong catalog engine, but order orchestration was a partner add-on, not native. Remains the recommended fallback if Vendor B contract negotiations fail.

Source: 05-phase-e-opportunities-and-solutions/vendor-evaluation.md

Diagram 17 — Vendor Evaluation Scorecard

8. Phase F — Migration Planning

8.1 Migration Roadmap

The roadmap runs four waves over 18 months, following a Wave 0 foundation. Wave 0 (months 1–3) stands up the ARB, ratifies the architecture principles, completes and contracts vendor selection, and establishes an integration inventory baseline — governance and vendor foundation with no customer-facing change yet. Wave 1 (months 3–7) deploys the API Gateway/Strangler Layer and builds the Legacy Billing and Legacy Network Inventory Adapters, plus Customer Domain Service v1 reconciling CRM/Billing customer records — legacy systems become API-addressable for the first time, the foundation for every subsequent wave. Wave 2 (months 6–12) brings the Product Catalog Platform live with mobile and broadband product definitions migrated in, the Order Orchestration Platform live for standard single-domain orders, and the Dual-Write Reconciliation Service live — the first automated, non-bundle order-to-activation flows go live and provisioning time begins dropping.

Wave 3 (months 10–16) is where the board's mandate is met: converged bundle products are defined and launched in the Product Catalog, cross-domain orchestration for bundles goes live, the Network-Slice Orchestration Adapter goes live, and the first 5G-slice product launches commercially. Wave 4 (months 15–18) completes RBAC/access-control uplift across all new components, fully populates and enforces the integration inventory, completes Phase H change management activities, and confirms success metrics from Phase A at the quarterly health review — handing the governance operating model to steady-state ARB cadence. Waves overlap intentionally where dependencies allow, to compress the timeline against the 18-month deadline. Every wave transition requires ARB sign-off, and every wave's go-live is additionally subject to the VP Billing & Revenue Systems sign-off requirement given billing's business-continuity exposure; roadmap variance exceeding 4 weeks against any wave triggers a mandatory ARB review, and variance exceeding 8 weeks against the Wave 3 slice-launch milestone escalates directly to the CTO and Board Technology Committee.

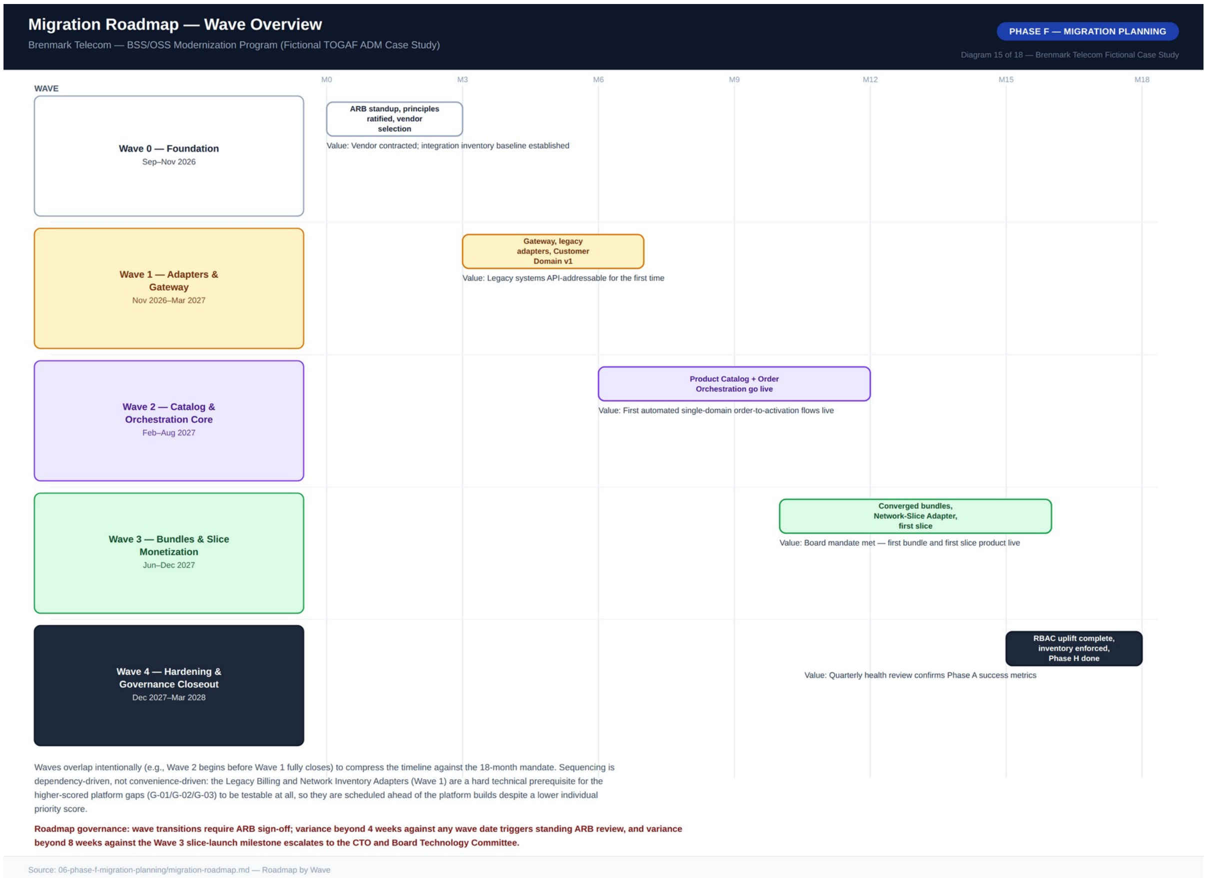


Diagram 15 — Migration Roadmap, Wave Overview

8.2 Transition Architectures

TOGAF Phase F distinguishes the as-is and to-be baselines from named intermediate states the organization deliberately operates in for a meaningful period, each with its own risk profile and governance obligations. Transition Architecture 1, "Gatewayed Legacy" (post-Wave 1, months 7–12): the API Gateway and both legacy adapters are live and enforced for new traffic, and the Customer Domain Service holds a reconciled customer record, but the Product Catalog and Order Orchestration platforms are not yet live — product definitions and order capture still run on legacy CRM/Billing modules, and some legacy-to-legacy batch sync still runs in parallel because the systems it updates haven't yet been superseded. This state validates that legacy systems can reliably respond to API traffic under production load before betting higher-value catalog and orchestration capability on that assumption. Its specific governance obligation: the integration inventory must show an explicit target retirement wave for every still-running legacy-to-legacy link — an unretired, unscheduled link found during this state is a Blocking finding, precisely because this is the state where such links are easiest to lose track of.

Transition Architecture 2, "Dual-Catalog Reconciliation" (post-Wave 2, months 12–18, overlapping Wave 3): the Product Catalog is authoritative and live for new single-domain sales, and standard orders auto-activate, but the legacy Billing system's internal product tables still exist and are still read by the billing engine's rating logic, kept consistent via the Dual-Write Reconciliation Service on a defined interval, not real-time. Converged bundles and slice products are being defined and tested but have not yet launched commercially — this is the immediate precursor to Wave 3's launch, not the launch state itself. Two definitions of "product" now concurrently exist in production, the specific condition P-01 is normally written to prevent, tolerated here only because it is time-boxed and governed under ADR-004. This state exits — and the legacy tables become formally deprecated — only once reconciliation monitoring shows sustained accuracy and lag within threshold across at least one full billing cycle following the Wave 3 bundle launch, a Phase G governance checkpoint rather than an automatic calendar event. Neither transition architecture is the program’s end state; both are states Brenmark actually operates in, in production, for months at a time, with governance obligations the to-be architecture itself does not carry.

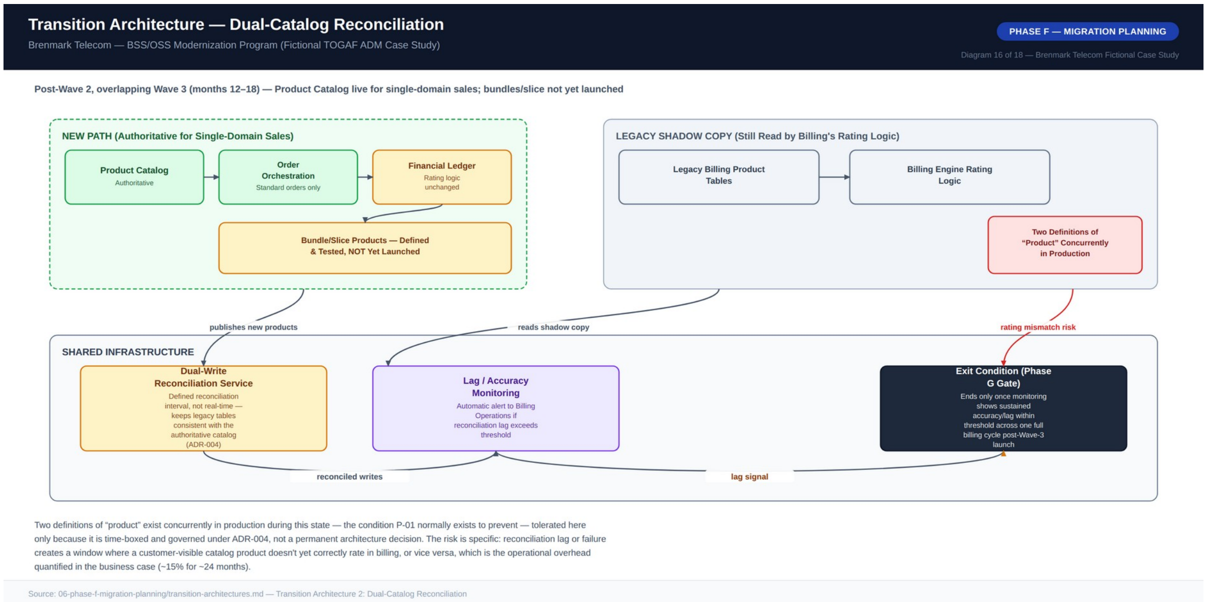


Diagram 16 — Transition Architecture, "Gatewayed Legacy"

9. Phase G — Implementation Governance

Phase G governs how the target architecture defined in Phases B–E is actually built, ensuring delivery teams stay within the approved architecture without turning the ARB into a bottleneck for routine engineering. Four mandatory compliance checkpoints apply across the delivery lifecycle: Contract Sign-Off (before build starts on any new solution building block or interface, gated by the ARB); Design Conformance Review (mid-build, checking the emerging design against the signed contract and applicable technology standards); Pre-Production Readiness Review (before go-live, checking NFR acceptance evidence, security sign-off, and any outstanding exceptions); and Post-Go-Live Health Check (a defined period after go-live, confirming the component performs against its committed NFRs in production, not just in test).

9.1 Non-Compliance Handling

The three non-compliance tiers described in Section 2.2 apply throughout delivery: Advisory findings are logged with a 60-day remediation window and do not block go-live; Must-Fix-Before-Production findings block the Pre-Production Readiness Review from passing until remediated and re-presented to the ARB; and Blocking findings — a P-01/P-06 violation with no approved exception, or an unregistered P-10 integration — halt deployment immediately, with escalation per the standard escalation path. An approved exception that expires without a logged renewal request automatically converts to a Blocking finding rather than continuing silently, closing the loop between the exceptions process (Section 6.2) and implementation governance.

9.2 Architecture Contracts

An Architecture Contract is the formal agreement between the ARB and a delivery team. The worked example is the Order Orchestration Platform Wave 2 contract, structured across nine standard sections: a scope statement; principles cited (P-02 Open API conformance, P-09 automated provisioning as default, P-11 binding non-functional requirements); applicable technology standards; disclosed exceptions (the Network Inventory SOAP activation-confirmation callback, with its 9-month expiry carried forward from Section 6.2); non-functional acceptance criteria (≤15 minutes for the 95th percentile of standard order-to-activation, ≥99.9% availability, ≤20% manual exception rate); interfaces and dependencies (the API Gateway, Legacy Billing Adapter, Legacy Network Inventory Adapter, and Customer Domain Service); EA function commitments (design review turnaround, checkpoint scheduling); a checkpoint schedule tied to program months 8, 11, and 12.5; and a sign-off section recording ARB approval. The contract explicitly states what happens when terms are missed: a missed non-functional acceptance criterion at Pre-Production Readiness Review blocks go-live until remediated and re-presented, not a waived pass, and a missed checkpoint date without an ARB-approved extension is itself escalated as a governance finding, not silently absorbed into the schedule.

10. Phase H — Change Management

An architecture that is technically correct but organizationally unadopted delivers none of the business case's projected value — the provisioning-time and manual-handoff reductions depend entirely on Customer Operations, Billing Operations, and Network Operations staff actually using the new capabilities as designed. Four staff populations are directly affected: Customer Operations CSRs move from manually re-keying orders across systems to capturing orders directly against the shared catalog, a high-impact workflow change; Billing Operations gains a new monitoring responsibility for dual-write reconciliation lag during the transition, also high impact; Network Operations moves from a manual work-order queue to API-driven activation confirmation for standard configurations, a high-impact change with a retained manual exception path; and Customer Operations Management gains new visibility into a unified customer/order view, a medium-impact change to reporting and oversight practice.

Training is sequenced to the migration waves: a foundational curriculum runs before Wave 1 begins (month 2), a lead-up curriculum ahead of Wave 2 (months 9–11) covers the new catalog and orchestration tooling before it becomes staff's primary system, a further lead-up curriculum ahead of Wave 3 (months 13–15) covers bundle and slice-specific workflows, and an ongoing knowledge base is maintained from Wave 2 onward. Communications run on a defined cadence by audience and channel, calibrated to avoid two failure modes: over-communicating a live-billing transition in a way that manufactures concern about a system that, if the transition architecture works as designed, should show customers no visible change; and under-communicating the 5–6 month Wave 1 period with no visible customer-facing progress, which the change management plan addresses directly since a silent period can otherwise be misread as the program stalling.

Adoption metrics reviewed alongside technical wave-gate criteria include the manual exception rate falling from 100% today to ≤20% by Wave 4 completion, staff confidence in the new tooling reaching ≥75% in post-training surveys, sustained knowledge-base usage, no increase in staff attrition attributable to the transition, and time-to-proficiency on the new tooling of ≤4 weeks for newly trained staff. Ownership of each change-management activity follows the Phase A RACI: VP Customer Operations is Responsible for the organizational change plan, with the Program Director Accountable and the CTO Consulted, ensuring change management carries the same governance rigor and its own budget line ($1.1M, per the business case) as the technical migration.

11. Architecture Decision Records

Five ADRs anchor the architecturally significant decisions in this program. Each follows the standard Status / Context / Decision / Alternatives Considered / Consequences / Stakeholders format and is cross-referenced from the architecture contracts and technology standards that depend on it.

ADR-001: Strangler-Fig Modernization vs. Full Rip-and-Replace

Modernize using a strangler-fig pattern: a TM Forum Open API–conformant gateway and a new Product Catalog and Order Orchestration layer are placed in front of the legacy billing, CRM, and network inventory systems, with legacy systems reached via API and capability migrated off incrementally rather than through a single cutover. A full BSS/OSS rip-and-replace was rejected because full replacement of a real-time charging engine for 6M subscribers is a multi-year undertaking incompatible with the 18-month mandate, concentrates all migration risk into one or a few large cutover events the business's risk appetite will not accept, and was modeled at 3–4x the strangler-fig approach's $18.4M capex with payback extending beyond the 3-year evaluation horizon. A greenfield BSS/OSS for new products only, with legacy left permanently untouched for existing subscribers, was also rejected: it does not solve the underlying fragmentation for the existing base, creates a second permanent BSS/OSS estate to operate indefinitely, and violates P-01 by design. Accepted trade-off: the strangler layer must maintain dual-write consistency between legacy product tables and the new catalog for an estimated 24 months (ADR-004), introducing reconciliation complexity and ~15% ongoing operational overhead, and the architecture carries two concurrent "views" of certain data during the transition, a condition P-01 normally disallows, tolerated only because it is explicitly time-boxed and governed. Stakeholders: ARB (approving body); VP Billing & Revenue Systems (primary risk owner, go-live sign-off); CTO and Board Technology Committee (accountable for the 18-month mandate outcome).

ADR-002: Product Catalog System of Record Placement

Designate the new Product Catalog Platform (Vendor B, Northolt BSS Cloud) as the sole system of record for product definitions across mobile, broadband, 5G-slice, and bundle products; the legacy billing system's internal product tables become a derived, reconciled copy via the Dual-Write Reconciliation Service (ADR-004), not an independent source of truth, with all sales channels and Order Orchestration consuming product data exclusively from the Product Catalog via TM Forum Open API. Designating the legacy billing product tables as the extended enterprise system of record was rejected: the billing product model was designed for mobile rating/charging, not bundles or 5G-slice SLA parameters, and extending it would entangle product-definition logic with financial-ledger logic Phase A protects from change, plus subject every catalog change to billing's slower regulatory change-control cycle. Designating the network inventory system as system of record was also rejected: its data model is oriented around network resources and capacity, not commercial product/pricing constructs, and it would put commercial pricing/bundling decisions under Network Engineering's change-control process rather than Product/Commercial ownership. Accepted trade-off: the billing engine's rating logic still reads from its own internal product tables — the Product Catalog is authoritative, not literally the only place product data exists — requiring the ADR-004 reconciliation mechanism and its ~15% transitional overhead, plus switching cost to migrate currently-siloed product definitions into the new catalog before Wave 2. Stakeholders: ARB (approving body, Head of Data & Analytics accountable for data governance); Head of Product/Commercial (primary business owner); VP Billing & Revenue Systems (consulted, given the reconciliation dependency).

ADR-003: TM Forum Open API Gateway Placement

Implement a single centralized API Gateway/Strangler Layer as the sole ingress and egress point between all new components, all channels, and all legacy systems; no new component integrates directly with a legacy system's native interface, every such integration is a thin adapter behind the same centralized gateway, which enforces authentication, schema validation, rate limiting, and routing uniformly. Distributed gateways, one per new component, were rejected: they reproduce, at smaller scale, exactly the integration fragmentation the program exists to eliminate, make P-10 unenforceable without a central audit point, and multiply the coordination burden of any legacy-system change. No dedicated gateway layer, with new components calling legacy native interfaces directly, was also rejected: it directly violates P-02 by baking legacy-specific translation logic into every consumer, and undermines P-04 reversibility since every future legacy retirement would require changing every consuming component rather than one adapter behind a stable contract. Accepted trade-off: the centralized gateway is a single point of failure for all order and catalog traffic if not built to its 99.95% availability target, raising the criticality and cost of the gateway component itself, and introduces a coordination bottleneck since every new integration must go through the centrally-governed onboarding process. Stakeholders: ARB (approving body); all SBB delivery teams (bound as a mandatory dependency); Head of InfoSec (central P-06 enforcement point for cross-system traffic).

ADR-004: Dual-Write Reconciliation Strategy for the Transition Period

Implement a scheduled, one-directional reconciliation service (the Dual-Write Reconciliation Service SBB) that reads authoritative product changes from the Product Catalog and applies corresponding updates to the legacy billing system's internal product tables on a defined interval, with active lag monitoring and automatic alerting to Billing Operations if lag exceeds threshold; legacy tables are explicitly a derived, non-authoritative copy with no write path back to the catalog, and the service is retired only once monitoring shows sustained accuracy and lag within threshold across at least one full billing cycle following the Wave 3 bundle launch. A true bidirectional dual-write with both systems co-authoritative was rejected: bidirectional consistency is a substantially harder distributed-systems problem, and it conflicts with P-01 in a way one-directional reconciliation does not, since bidirectional dual-write has no single authority at all. Real-time synchronous replication was also rejected for the transition period: it would require the legacy billing system to accept synchronous inbound calls on every catalog change, risking real-time charging performance, and it raises reconciliation-service outage stakes from "growing, monitored lag" to "hard synchronous coupling failure" for a component explicitly intended to be temporary. Accepted trade-off: this is the specific mechanism generating the ~15% operational overhead for an estimated 24 months, and it introduces a real, monitored, bounded risk window where a catalog-visible product has not yet propagated to billing's rating tables, requiring active operational monitoring (a Billing Operations role change, Phase H) rather than a fire-and-forget process. Stakeholders: ARB (approving body — reviewed under P-01 but deemed a compliant, time-boxed implementation of P-01 rather than an exception requiring unanimous approval, a distinction the ARB minutes record explicitly); VP Billing & Revenue Systems (primary operational risk owner); Billing Operations staff (direct workflow impact, Phase H).

ADR-005: Network-Slice Orchestration Integration Approach

Build a dedicated Network-Slice Orchestration Adapter as an in-house solution building block, integrating the chosen Order Orchestration Platform (Vendor B) with the network layer's slice-capable orchestration tooling, coordinated directly with Network Engineering's parallel 5G capital program rather than depending on the commercial BSS/OSS vendor to provide this integration natively — scoped as a genuine build because no evaluated vendor's out-of-box slice orchestration integration adequately fit Brenmark's specific network equipment vendor mix. Selecting Vendor D (Ferro, strongest on slice orchestration at 5/5) as the primary platform instead of Vendor B was rejected: Vendor D scored weakest of all four on converged-bundle commerce modeling and Open API conformance, the two highest-weighted criteria, and the business case shows converged-bundle revenue as the larger near-term value driver, making a platform optimized narrowly for slice orchestration a poor fit for the program's overall value proposition. Adopting Vendor D's slice-orchestration module as a bolt-on alongside the Vendor B platform was also rejected: it introduces a second commercial vendor relationship into the critical path for the highest-priority mandate item, and creates exactly the kind of vendor-to-vendor point-to-point integration P-10 is designed to prevent, since it would need to route through the centralized gateway anyway. Accepted trade-off: Brenmark carries the engineering cost and delivery risk of an in-house build (estimated 105 person-months) rather than a vendor's pre-built integration, on the critical path for the board's mandate (Wave 3), coordinated with a parallel Network Engineering capital program whose timeline is a dependency this architecture does not fully control. Stakeholders: ARB (approving body); VP Network Engineering (primary technical dependency owner); CTO and Board Technology Committee (schedule variance on this adapter is a mandatory escalation trigger, since it directly determines mandate achievability).

Appendix A — Diagram Catalog Index

The full reference diagram catalog for this program comprises 18 diagrams spanning the Preliminary Phase through Phase F, delivered as high-resolution PNG files alongside this document.

No.	Diagram Title	ADM Phase
01	Architecture Vision One-Pager	Phase A
02	Stakeholder Power / Interest Map	Phase A
03	Architecture Governance Organization	Preliminary
04	Business Capability Map, Maturity Heat Map	Phase B
05	Business Process Flow, As-Is (Order-to-Activation)	Phase B
06	Business Process Flow, To-Be (Order-to-Activation)	Phase B
07	Business Footprint Diagram	Phase B
08	Application Communication Diagram, As-Is	Phase C
09	Application Communication Diagram, To-Be	Phase C
10	Data Flow Diagram, To-Be	Phase C
11	Data Ownership / Domain Diagram	Phase C
12	Reference Architecture Pattern	Phase D
13	Technology Standards Stack	Phase D
14	Solution Building Blocks, Buy / Build	Phase E
15	Migration Roadmap, Wave Overview	Phase F
16	Transition Architecture, "Gatewayed Legacy"	Phase F
17	Vendor Evaluation Scorecard	Phase E
18	Business Case Financial Summary	Phase A

Appendix B — Disclaimer

This entire document — Brenmark Telecom, all named individuals and roles, all vendor names (Meridian Commerce Suite, Northolt BSS Cloud, Cascade Digital Commerce, Ferro OSS/BSS Platform), and every dollar figure, metric, and date — is a fictional case study constructed for architecture-practice demonstration purposes. Figures are designed to be internally consistent and arithmetically sound so that the reasoning pattern is transparent, not to represent any real telecommunications operator's actual data. No inference should be drawn about any real company, product, or individual.